Experiment 5

Title: Implementation of Rule-Based Part of Speech (POS) Tagging

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Aim

To implement **rule-based Part of Speech (POS) tagging** for identifying the grammatical categories of words in a given text.

Objectives

1. To understand the concept of Part of Speech tagging in text processing.
2. To study rule-based methods used for POS tagging.
3. To apply rule-based POS tagging on a given sentence or text.

Theory

1. Introduction to Part of Speech Tagging
Part of Speech (POS) tagging is a technique used in **Natural Language Processing (NLP)** to assign grammatical categories to each word in a sentence. These categories include nouns, verbs, adjectives, adverbs, pronouns, and others. POS tagging helps computers understand the structure and meaning of a sentence.

2. Rule-Based POS Tagging
Rule-based POS tagging uses **predefined linguistic rules** to assign tags to words. These rules are created based on grammar patterns, word endings, or the position of a word in a sentence. For example, words ending with "-ing" are often tagged as verbs, and words following determiners like "the" are usually nouns.

3. Working of Rule-Based Tagging
In this method, the system first assigns an initial tag to each word using a dictionary or lexicon. Then a set of grammatical rules is applied to correct or refine the tags. These rules consider the context of surrounding words to determine the correct part of speech.

4. Applications of POS Tagging
POS tagging is widely used in **text analysis, machine translation, speech recognition, and information extraction**. It helps improve the understanding of sentence structure and plays an important role in many NLP tasks.

Conclusion

In this experiment, rule-based Part of Speech tagging was studied and implemented. This method assigns grammatical tags to words using predefined linguistic rules. It helps identify the role of each word in a sentence and improves text understanding. POS tagging is an important step in many natural language processing applications. Therefore, rule-based tagging is useful for analyzing and processing textual data.